COMPUTER VISION

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Class Test

1. **Image Enhancement in the Spatial Domain**

Definition

Image Enhancement in the spatial domain improves the visual quality of an image by directly modifying the intensity values of its pixels

The general expression is :

$$g(x, y) = T\,[f(x, y)]$$

Where :

* $f(x, y)$ = input image
* $g(x, y)$ = Enhanced Image
* T = Transformation function

(A) Gray-level Transformations

These modify pixel intensity values

Types

1. Image Negative
    * Formula : $s = L - 1 - r$

* Highlights details in dark regions

2. log transformation

* Formula $s = c \log(1+r)$

* Expands dark pixels and compresses bright pixels.

3. power-law (Gamma) Transformation

* Formula : $S = cr^{\gamma}$

* corrects brightness

4. contrast stretching

* increases contrast between dark and bright regio

Suitability

* Medical images

* Satellite images

* low-light photography

(B) Histogram processing

A histogram shows the distribution of gray levels.

Techniques

Histogram Equalization

* improves global contrast

* Spreads intensity values uniformly

Histogram Equalization

* improves global contrast

Suitability

* poor contrast images
* Medical Image enhancement
* Remote sensing

(c) Spatial Filtering

uses a mask / kernel over neighboring pixels.

smoothing Filters

* Mean Filter
* Gaussian Filter
* Median Filter

purpose

* Remove noise

sharpening Filters

* Laplacian Filter
* Sobel Filter
* unsharp Masking

Purpose

* Enhance edges

Advantages

* Simple implementation
* Fast computation
* improves visibility
* Enhances image quality

Limitations

* May amplify noise
* Excessive sharpening creates artifacts

Conclusion

Gray level transformations improves brightness histogram processing improves contrast, and Spatial filtering removes noise or sharpens details, making images suitable for further processing

2. Image smoothing and image sharpening

Image smoothing

Image smoothing reduces noise and small intensity variations using low-pass filters

Filters ;

* Mean Filter
* Gaussian Filter
* Median Filter

Advantages

* Removes noise
* produces smooth images
* improves preprocessing

Image sharpening:

Image sharpening enhances edges and details using high pass filters

Filters :

* Laplacian
* Sobel
* Prewitt

Advantages

* Highlights edges
* improves object recognition
* Make image clearer

Difference :-

Image smoothing	Image sharpening
Removes noise	Enhances edges
uses low-pass filters	uses high-pass filters
Blurs the image	Sharpens the image

Applications :- smoothing is used before segmentation while sharpening is used for feature extraction and object recognition.

3. Image Enhancement in Frequency Domain.

Definition :- Fr

Image Enhancement in the frequency domain improves image quality by modifying frequency components using the Fourier transform (FFT)

process :-

Image → FFT → Apply LPF/HPF → IFFT → Enhanced image

Low-pass Filter (LPF):

* passes low frequencies
* Removes noise and smooths the image.
* used in medical and satellite images

High pass Filter (HPF):

* passes high Frequencies
* Enhances edges and sharpens the image.
* used in edge detection and object recognition

Block diagram:

Input image

↓

Fourier Transform (FFT)

↓

Apply Frequency transform

(LPF / HPF)

↓

Inverse Fourier Transform

↓

Enhanced image.

4. Image segmentation

Definition :

Image segmentation divides an image into meaningful regions or objects.

Technique	use
point Detection	Detects isolated pixels
Line Detection	Detects straight lines
Edge Detection	Detects object ~~boundary~~
Thresholding	seperates object from background.
Region based	Groups similar.

Best Technique :

* Thresholding - simple images
* Edge Detection + Region Based - complex images.

Edge Detection, Edge linking & Boundary Detection

Edge Detection:-

Finds object edges by detecting intensity changg

Edge linking :-

connects broken edges to form continuous boundaries.

Boundary Detection:

Identifies the outer boundary of an object.

Applications :

* object recoginition
* Medical imaging
* Crack / defect detection
* Face recogintion.

Flow

Image

↓

Edge detection

↓

Edge linking

↓

Boundary Detection

↓

Object Recognition